HEALTH CARE PROJECT

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Introduction:

We all know that Health care is very important domain in the market. It is directly linked with the life of the individual; hence we have to be always be proactive in this particular domain. Money plays a major role in this domain, because sometime treatment becomes super costly and if any individual is not covered under the insurance then it will become a pretty tough financial situation for that individual. The companies in the medical insurance also want to reduce their risk by optimizing the insurance cost, because we all know a healthy body is in the hand of the individual only. If individual eat healthy and do proper exercise the chance of getting ill is drastically reduced.

Objective: The objective of this exercise is to build a model, using data that provide the optimum insurance cost for an individual. You have to use the health and habit related parameters for the estimated cost of insurance

Data Description:

Variable	Definition
applicant_id	Applicant unique ID
years_of_insurance_with_us	Since how many years customer is taking policy from the same company only
regular_checkup_lasy_year	Number of times customers has done the regular health check up in last one year
Occupation	Occupation of the customer
visited_doctor_last_1_year	Number of times customer has visited doctor in last one year
cholesterol_level	Cholesterol level of the customers while applying for insurance

daily_avg_steps	Average daily steps walked by customers
age	Age of the customer
heart_decs_history	Any past heart diseases
other_major_decs_history	Any past major diseases apart from heart like any operation
Gender	Gender of the customer
avg_glucose_level	Average glucose level of the customer while applying the insurance
bmi	BMI of the customer while applying the insurance
smoking_status	Smoking status of the customer
Year_last_admitted	When customer have been admitted in the hospital last time
Location	Location of the hospital
weight	Weight of the customer
covered_by_any_other_company	Customer is covered from any other insurance company
Alcohol	Alcohol consumption status of the customer
exercise	Regular exercise status of the customer
weight_change_in_last_one_year	How much variation has been seen in the weight of the customer in last year

fat_percentage	Fat percentage of the customer while applying the insurance
insurance_cost	Total Insurance cost

Data Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   applicant_id                    25000 non-null  int64
 1   years_of_insurance_with_us      25000 non-null  int64
 2   regular_checkup_lasy_year       25000 non-null  int64
 3   adventure_sports                25000 non-null  int64
 4   Occupation                      25000 non-null  object
 5   visited_doctor_last_1_year      25000 non-null  int64
 6   cholesterol_level               25000 non-null  object
 7   daily_avg_steps                 25000 non-null  int64
 8   age                             25000 non-null  int64
 9   heart_decs_history              25000 non-null  int64
 10  other_major_decs_history        25000 non-null  int64
 11  Gender                          25000 non-null  object
 12  avg_glucose_level               25000 non-null  int64
 13  bmi                             24010 non-null  float64
 14  smoking_status                  25000 non-null  object
 15  Year_last_admitted              13119 non-null  float64
 16  Location                        25000 non-null  object
 17  weight                          25000 non-null  int64
 18  covered_by_any_other_company    25000 non-null  object
 19  Alcohol                         25000 non-null  object
 20  exercise                        25000 non-null  object
 21  weight_change_in_last_one_year  25000 non-null  int64
 22  fat_percentage                  25000 non-null  int64
 23  insurance_cost                  25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table 1: dataset details

Note:

- The data contains 25000 observations and 24 variables.
- Data's were missing in the dataset especially in bmi and Year_last_admitted.
- The variables contain continuous data are applicant_id, years_of_insurance_with_us, regular_checkup_lasy_year visited_doctor_last_1_year, daily_avg_steps, age, heart_decs_history, other_major_decs_history, avg_glucose_level, Bmi, Year_last_admitted,

Weight, weight_change_in_last_one _year, fat_percentage and insurance_cost made.

- The data types of the variables present are integer, float and object.
- There are 16 numeric (*int and *float type) and 8 object type.

Statistical summary of the data:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
applicant_id	25000.0	NaN	NaN	NaN	17499.5	7217.022701	5000.0	11249.75	17499.5	23749.25	29999.0
years_of_insurance_with_us	25000.0	NaN	NaN	NaN	4.08904	2.606612	0.0	2.0	4.0	6.0	8.0
regular_checkup_last_year	25000.0	NaN	NaN	NaN	0.77368	1.199449	0.0	0.0	0.0	1.0	5.0
adventure_sports	25000.0	NaN	NaN	NaN	0.08172	0.273943	0.0	0.0	0.0	0.0	1.0
Occupation	25000	3	Student	10169	NaN	NaN	NaN	NaN	NaN	NaN	NaN
visited_doctor_last_1_year	25000.0	NaN	NaN	NaN	3.1042	1.141663	0.0	2.0	3.0	4.0	12.0
cholesterol_level	25000	5	150 to 175	8763	NaN	NaN	NaN	NaN	NaN	NaN	NaN
daily_avg_steps	25000.0	NaN	NaN	NaN	5215.88932	1053.179748	2034.0	4543.0	5089.0	5730.0	11255.0
age	25000.0	NaN	NaN	NaN	44.91832	16.107492	16.0	31.0	45.0	59.0	74.0
heart_decs_history	25000.0	NaN	NaN	NaN	0.05464	0.227281	0.0	0.0	0.0	0.0	1.0
other_major_decs_history	25000.0	NaN	NaN	NaN	0.09816	0.297537	0.0	0.0	0.0	0.0	1.0
Gender	25000	2	Male	16422	NaN	NaN	NaN	NaN	NaN	NaN	NaN
avg_glucose_level	25000.0	NaN	NaN	NaN	167.53	62.729712	57.0	113.0	168.0	222.0	277.0
bmi	24010.0	NaN	NaN	NaN	31.393328	7.876535	12.3	26.1	30.5	35.6	100.6
smoking_status	25000	4	never smoked	9249	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Year_last_admitted	13119.0	NaN	NaN	NaN	2003.892217	7.581521	1990.0	1997.0	2004.0	2010.0	2018.0
Location	25000	15	Bangalore	1742	NaN	NaN	NaN	NaN	NaN	NaN	NaN
weight	25000.0	NaN	NaN	NaN	71.61048	9.325183	52.0	64.0	72.0	78.0	96.0
covered_by_any_other_company	25000	2	N	17418	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Alcohol	25000	3	Rare	13752	NaN	NaN	NaN	NaN	NaN	NaN	NaN
exercise	25000	3	Moderate	14638	NaN	NaN	NaN	NaN	NaN	NaN	NaN
weight_change_in_last_one_year	25000.0	NaN	NaN	NaN	2.51796	1.690335	0.0	1.0	3.0	4.0	6.0
fat_percentage	25000.0	NaN	NaN	NaN	28.81228	8.632382	11.0	21.0	31.0	36.0	42.0
insurance_cost	25000.0	NaN	NaN	NaN	27147.40768	14323.691832	2468.0	16042.0	27148.0	37020.0	67870.0

Table 2: Statistical data details

- The number of years_of_insurance_with_us is minimum of 0 and the maximum of 8 years.
- The visited_doctor_last_1_year of minimum is 0 and maximum is 12.
- The 50% of the average glucose level is 168 and maximum value is 277.
- The weight_change_in_last_one_year has the maximum value of 6.
- The fat_percentage of the maximum value is 42.
- The target variable is insurance_cost with the maximum value of 67870.0

Missing values in Dataset:

- There are missing values in dataset.
- There are no duplicates values in dataset.

BEFORE TREATMENT:

applicant_id	0
years_of_insurance_with_us	0
regular_checkup_last_year	0
adventure_sports	0
Occupation	0
visited_doctor_last_1_year	0
cholesterol_level	0
daily_avg_steps	0
age	0
heart_decs_history	0
other_major_decs_history	0
Gender	0
avg_glucose_level	0
bmi	990
smoking_status	0
Year_last_admitted	11881
Location	0
weight	0
covered_by_any_other_company	0
Alcohol	0
exercise	0
weight_change_in_last_one_year	0
fat_percentage	0
insurance_cost	0
dtype: int64	

Table 3: Missing value before treatment data

- Missing values found in the bmi and Year_last_admitted.
- For float values we used median values.

AFTER TREATMENT:

applicant_id	0
years_of_insurance_with_us	0
regular_checkup_last_year	0
adventure_sports	0
Occupation	0
visited_doctor_last_1_year	0
cholesterol_level	0
daily_avg_steps	0
age	0
heart_decs_history	0
other_major_decs_history	0
Gender	0
avg_glucose_level	0
bmi	0
smoking_status	0
Year_last_admitted	0
Location	0
weight	0
covered_by_any_other_company	0
Alcohol	0
exercise	0
weight_change_in_last_one_year	0
fat_percentage	0
insurance_cost	0
dtype: int64	

Table 4: Missing value after treatment data

A. UNIVARIATE ANALYSIS
Continuous variable data

Years_of_insurance_with_us:

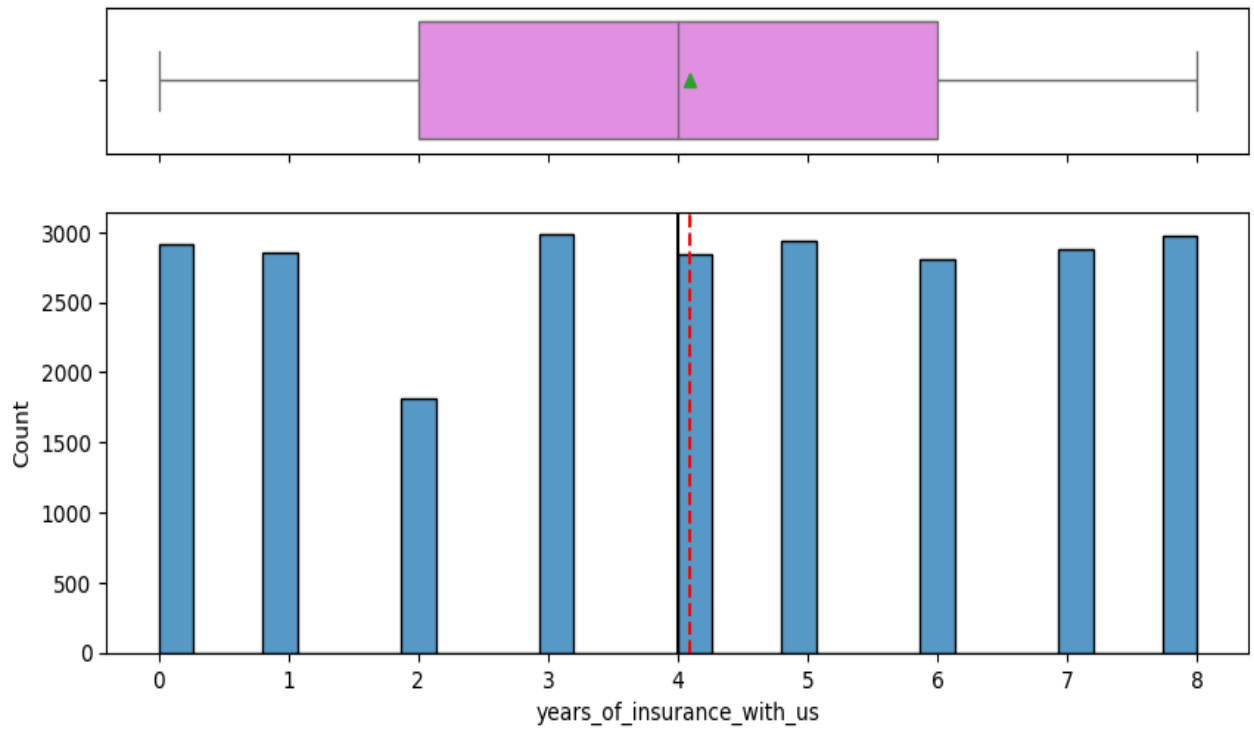

Figure 1: Boxplot and Histplot for Years_of_insurance_with_us

- The figure represents that the years_of_insurance starts from the minimum value of 0 to maximum value of 8 years.

Regular_checkup_last_year:

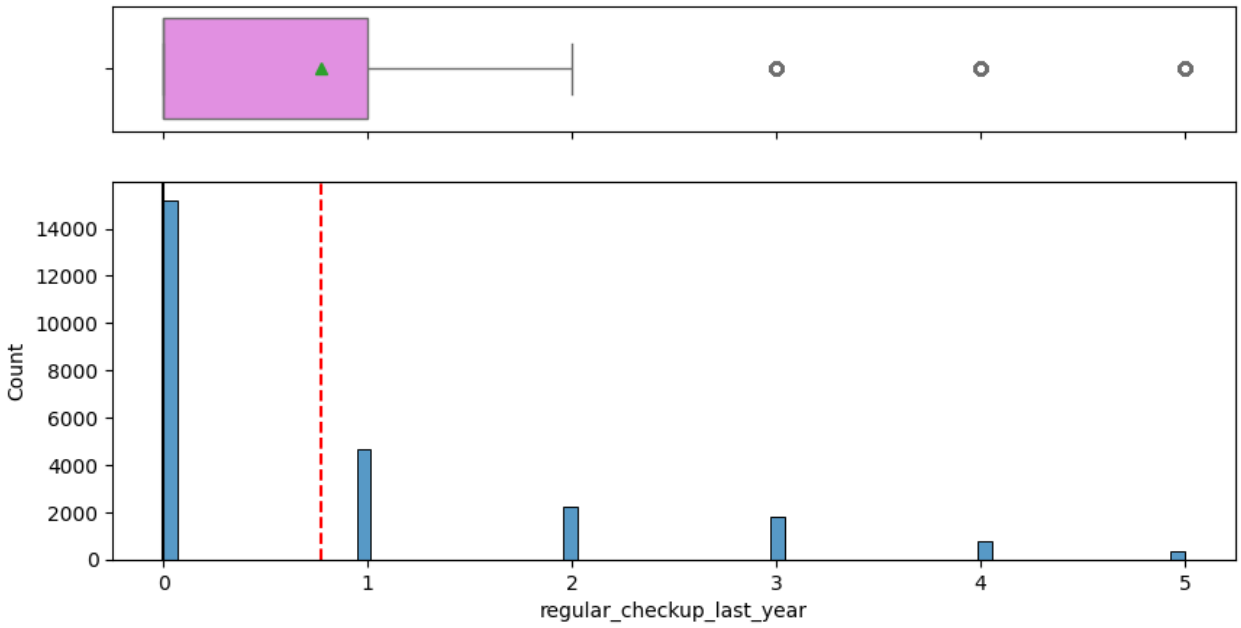

Figure 2: Boxplot and Histplot for Regular_checkup_last_year

- The figure represents that the data has minimum value of 0 and the maximum of 5.
- The nearly 80 % of the customer hasn't done regular_checkup_last_year and only few people do regular_checkup_last_year.

Adventure_sports:

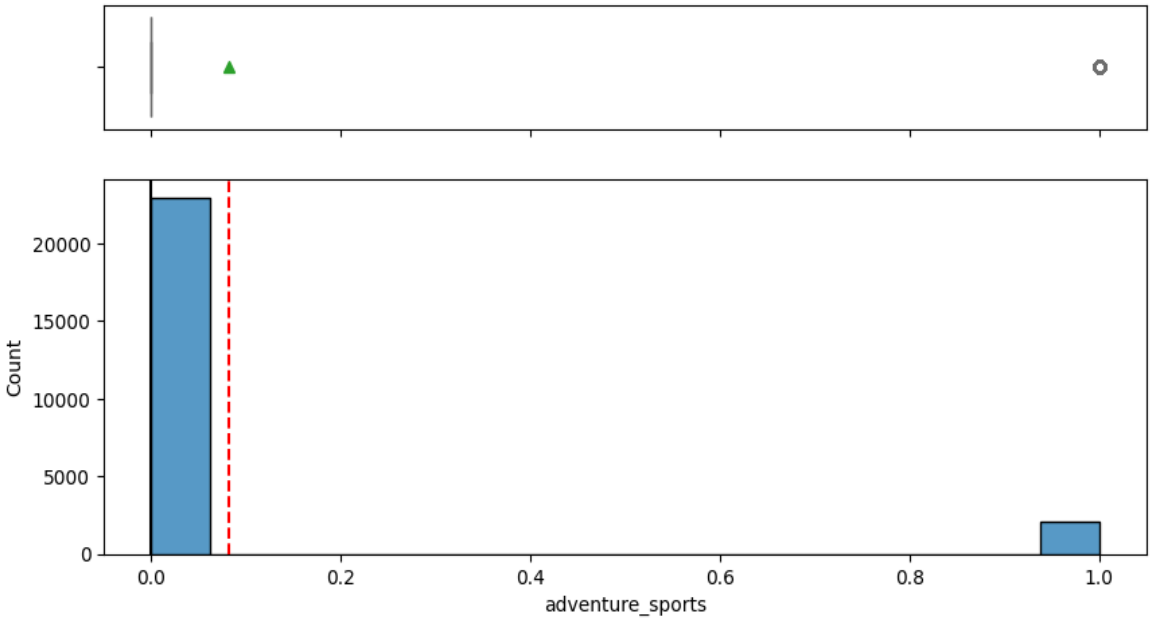

Figure 3: Boxplot and Histplot for adventure_sports

- The figure represents that minimum of 0 and the maximum of 1.

Visited_doctor_last_1_year:

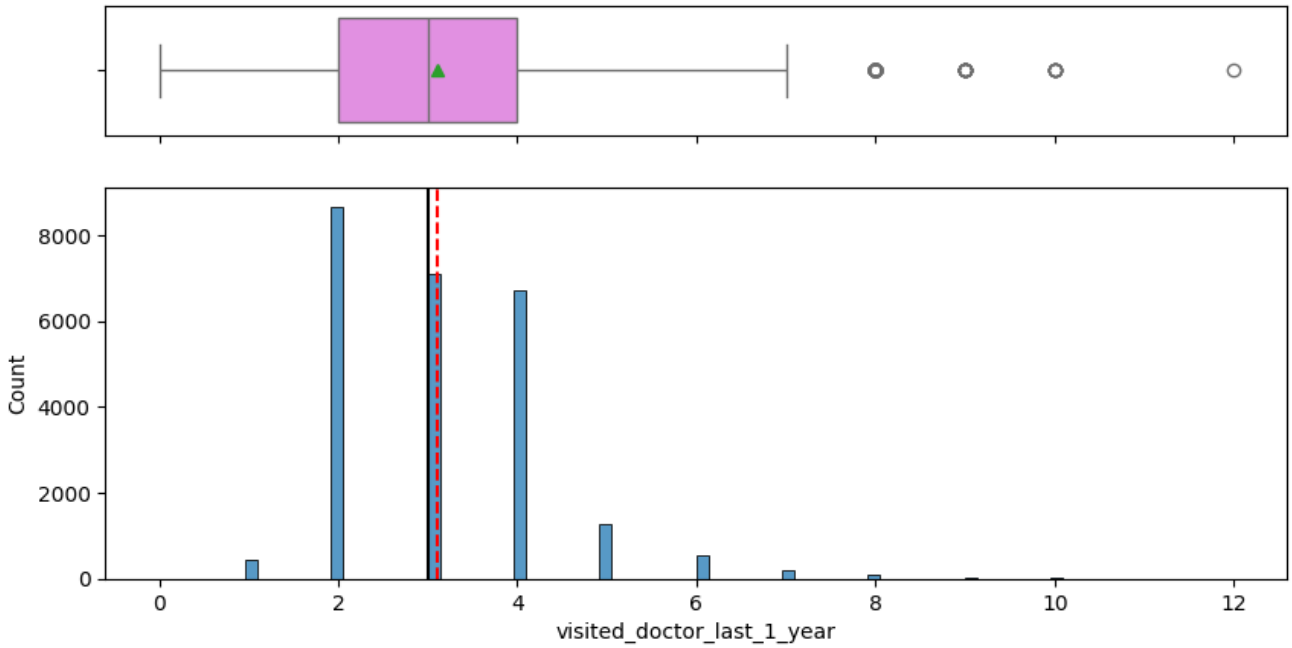

Figure 4: Boxplot and Histplot for Visited_doctor_last_1_year

- The figure represents that the Number of customers visited_doctor_last_1_year was 1 to 9.
- The minimum value is 1 and the maximum value is 10.
- This explains that the visited_doctor_last_1_year 50 % was only 7500.

Daily_avg_steps:

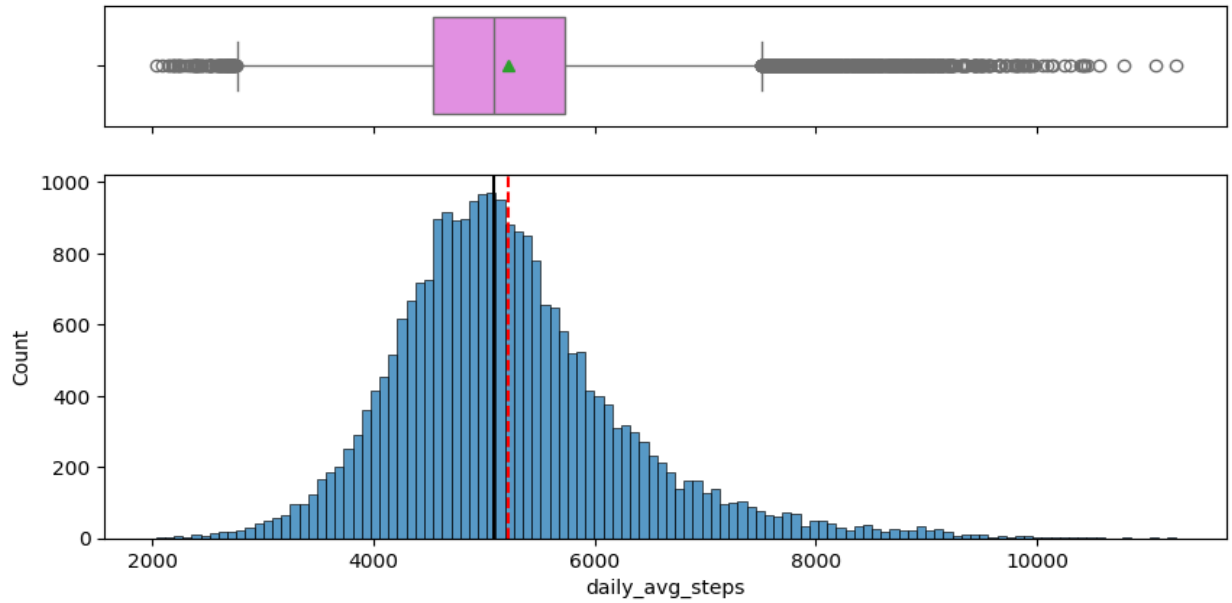

Figure 5: Boxplot and Histplot for Daily_avg_steps

- The figure represents that the daily_avg_steps have many outliers.
- This variable is heavily right skewed distribution and the mean value is 5000.

Age:

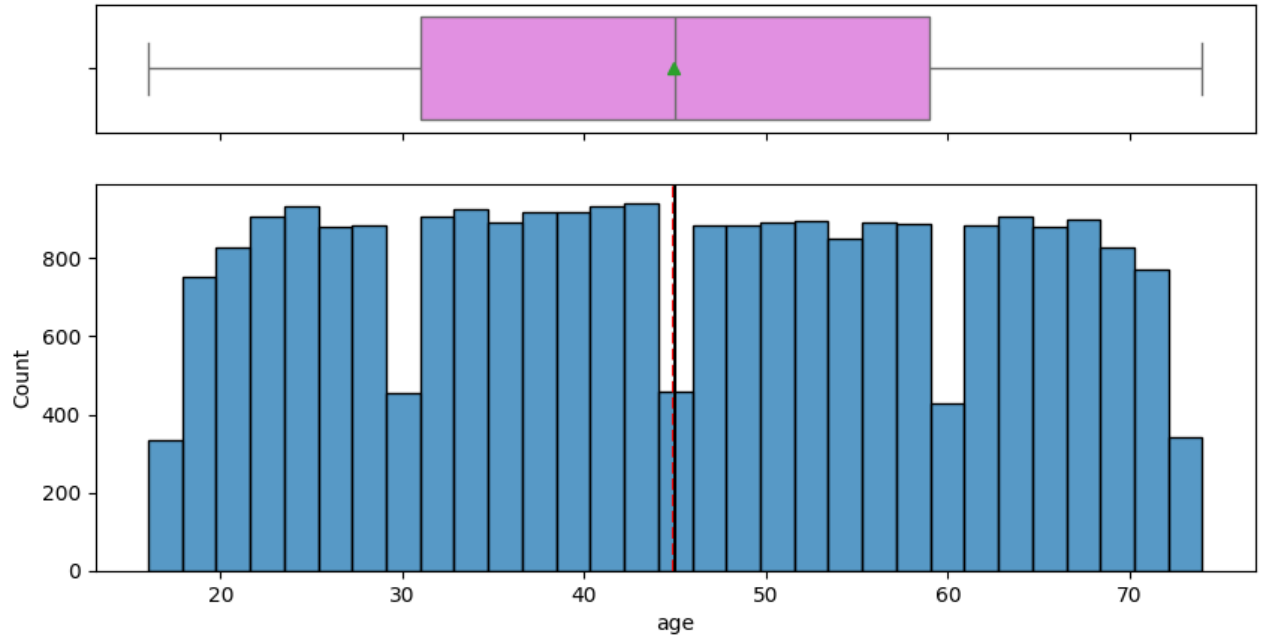

Figure 6: Boxplot and Histplot for Age

Central Tendency:

- Median ≈ 45 years
- Mean ≈ 45 years
- Spread: Age ranges approximately from 15 to 75 years.
- Distribution: Nearly uniform with some multimodal dips (at ~30 and ~60).
- Outliers: None detected.
- Skewness: Minimal (distribution is symmetric).

Heart_decs_history:

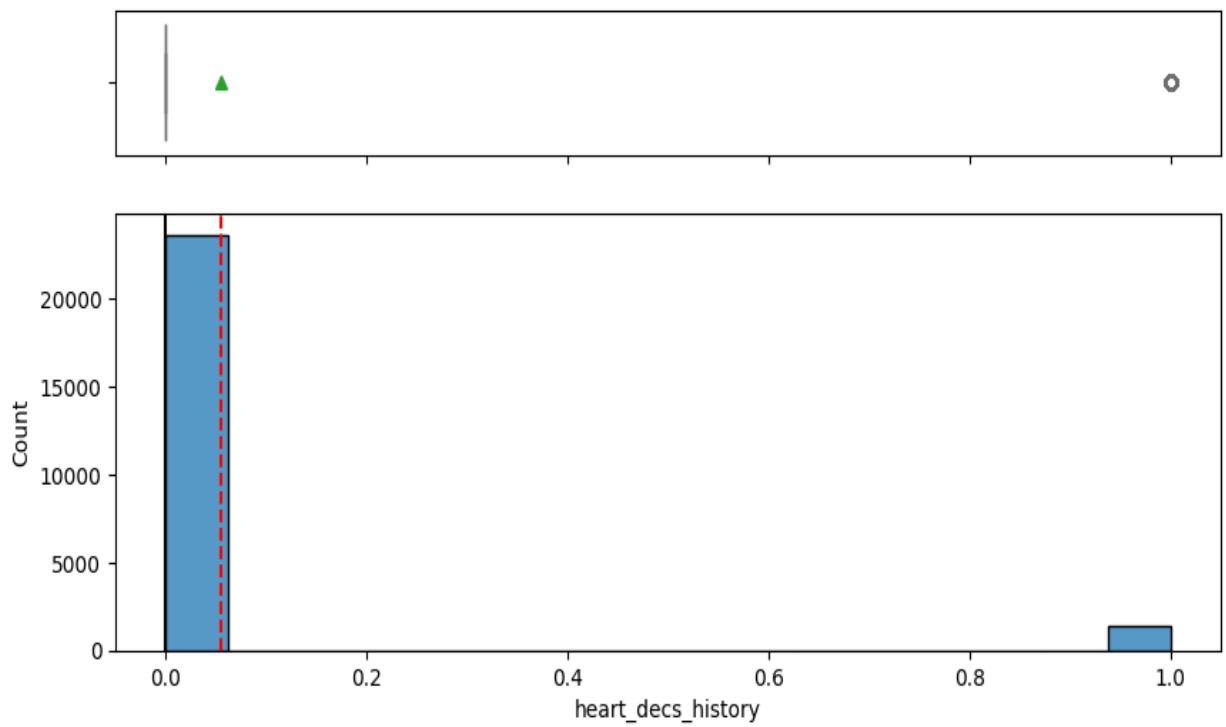

Figure 7: Boxplot and Histplot for Heart_decs_history

Boxplot Analysis (Top):

- A green triangle marks the mean, slightly above 0, indicating most people fall into category 0.
- The value 1 is treated as an outlier, shown as a separate point (circle).
- The box itself is very narrow around 0, confirming very low variance and a high concentration at 0.

Interpretation:

- Highly right-skewed data.

- The value 1 is infrequent and seen as an extreme value.

Histogram Analysis (Bottom)

- The bar for 0 is significantly taller (likely >22,000 instances).
- The bar for 1 is much shorter (few hundred or low thousands).
- A red dashed vertical line shows the mean, located close to 0.

Interpretation:

- The vast majority of people do not engage in adventure sports.
- Very few records are marked as 1, indicating adventure sports participation.
- This is a class-imbalanced variable.

Other_major_decs_history:

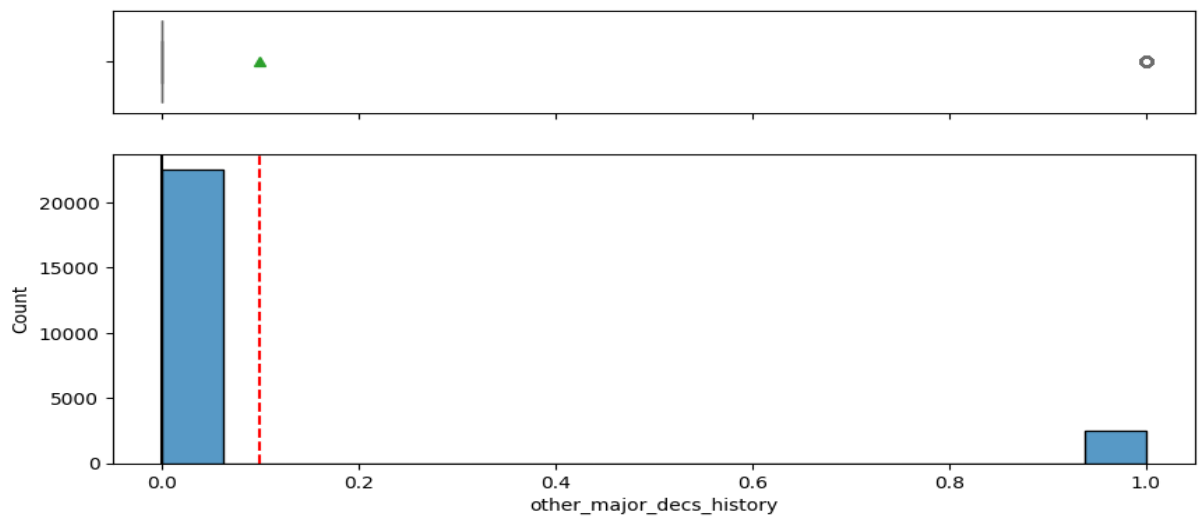

Figure 8: Boxplot and Histplot for Other_major_decs_history

• The data is highly skewed toward 0.

- The presence of 1 is rare and seen as an outlier.
- The bar at 0 is very tall, indicating the majority of data points fall here (~23,000).
- The bar at 1 is very short, indicating few observations have a history (~2,000 or less).
- A red dashed line indicates the mean (between 0 and 1, closer to 0).

Interpretation:

- Class imbalance is significant — most individuals have no major decision history (0).
- Very few individuals have a 1 in this column.

Avg_glucose_level:

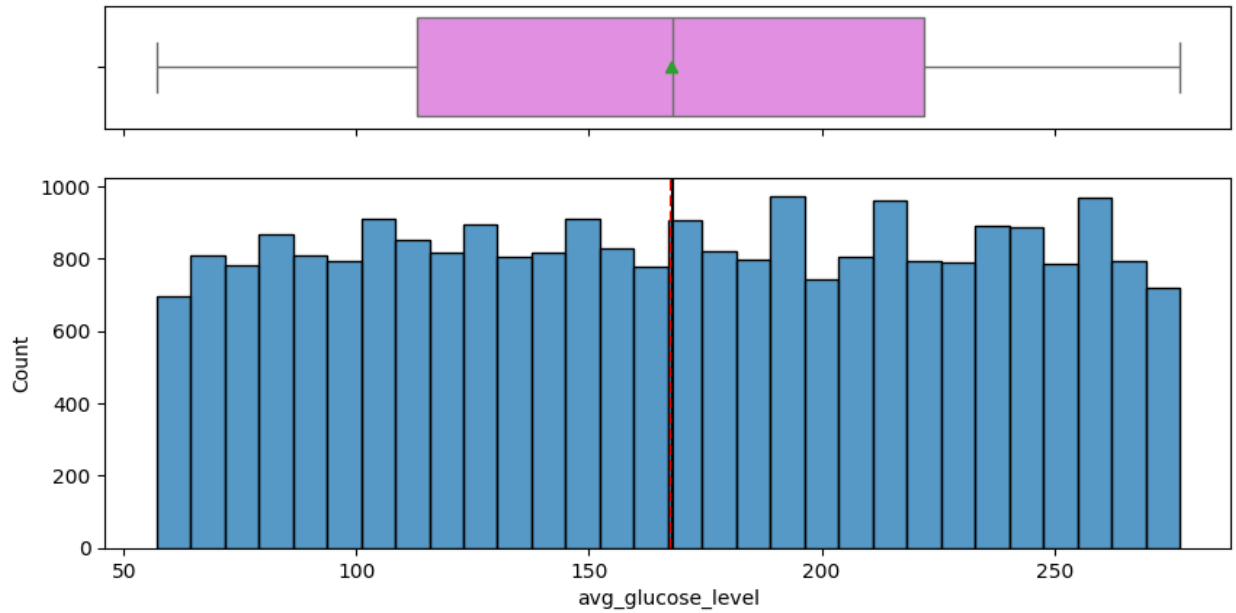

Figure 9: Boxplot and Histplot for Avg_glucose_level

Interpretation:

- Data is fairly symmetrical with no strong skewness.
- No outliers suggest the data is clean and consistent.
- Both the mean and median are almost aligned — indicates normal-like distribution.
- The data is evenly spread across the range.
- There is no heavy concentration in one area, and no long tails or skew.
- The distribution appears well-balanced and suitable for modeling without transformation.

Bmi:

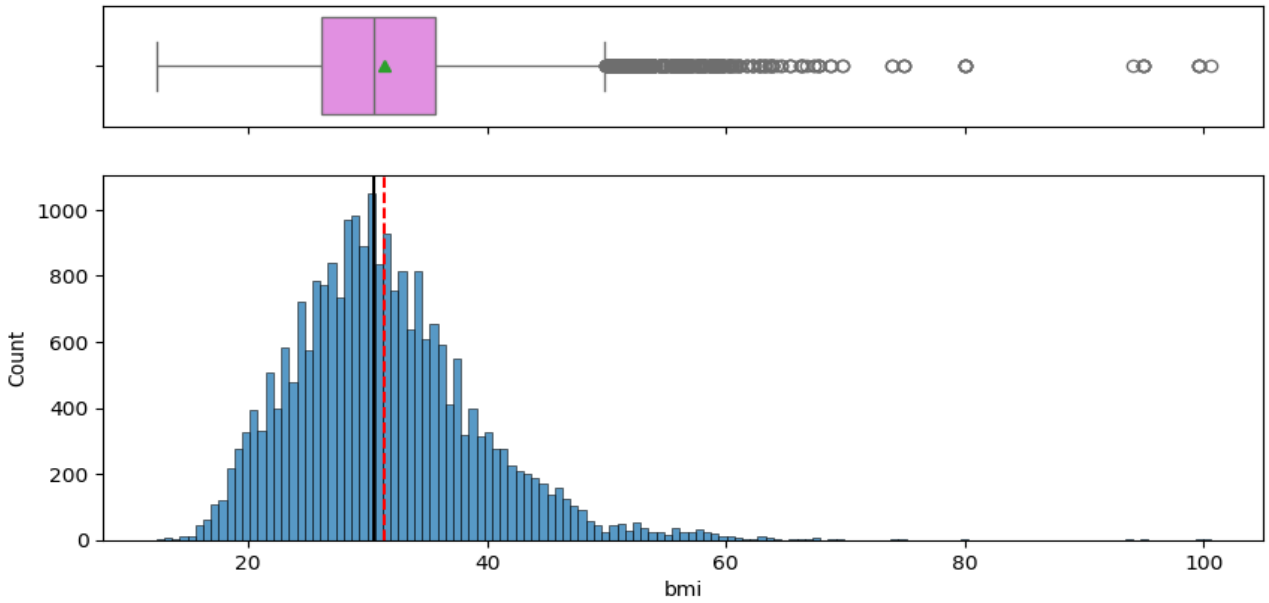

Figure 10: Boxplot and Histplot for Bmi

• The figure explains that the bmi has outliers and the data is heavily right skewed distribution.

• The figure explains that the dataset for workers_num has the mean value of 35.

Year_last_admitted:

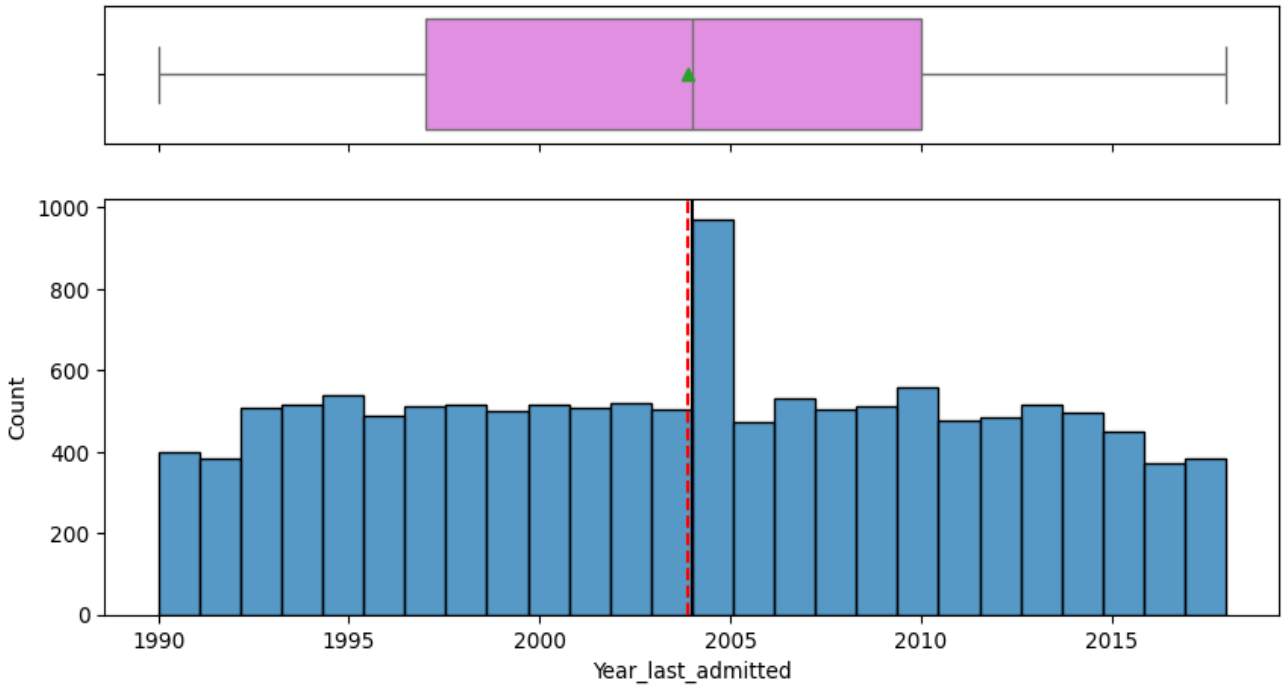

Figure 11: Boxplot and Histplot for Year_last_admitted

The distribution is bimodal-like, with slight peaks:

- Around 1994–1996
- A sharp rise near 2004
- The red dashed line marks the median year, approximately 2004.
- Counts range from ~380 to nearly 1,000 per year bin.

Weight:

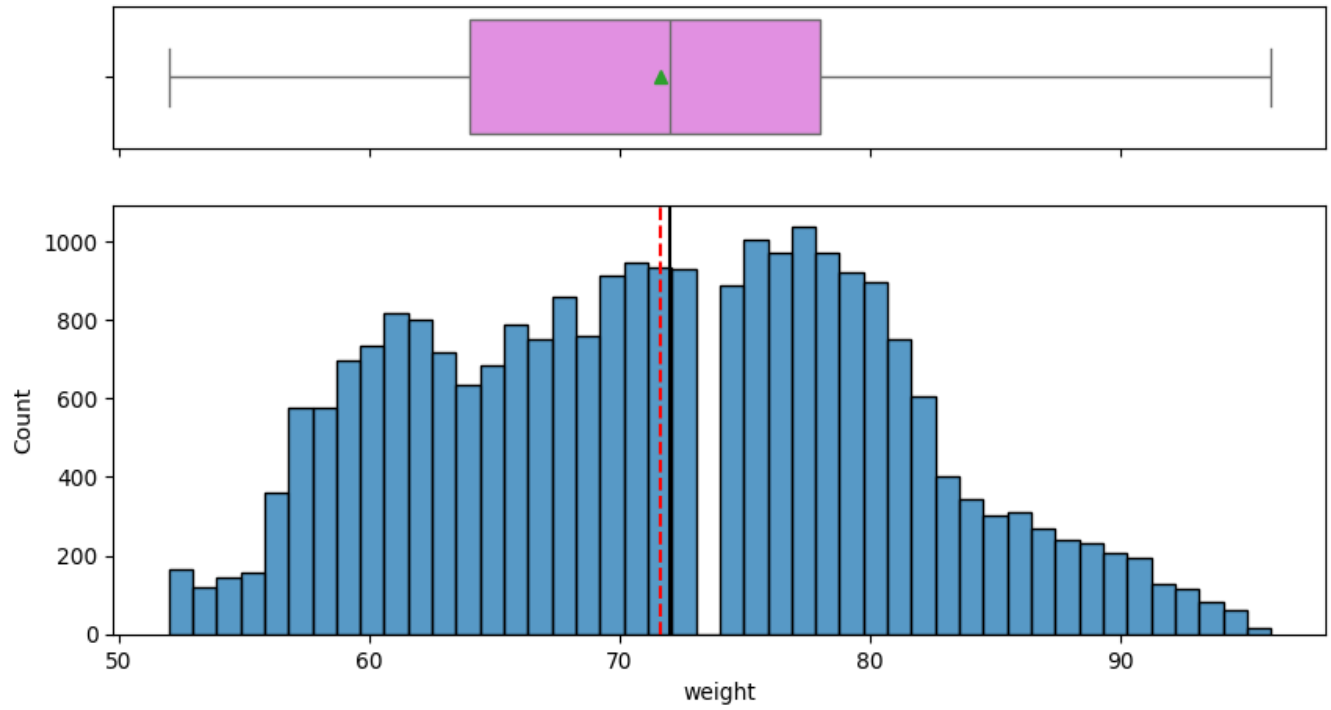

Figure 12: Boxplot and Histplot for Weight

• Boxplot Insights:

- The weight distribution is spread approximately from ~50 to ~95 kg.
- The box (IQR) spans approximately 60 to 80 kg, with the median (black line) slightly below 72 kg.
- A green triangle marks the mean, just to the right of the median, suggesting slight right skewness.
- No extreme outliers are visible.

Histogram Insights:

- The distribution is bimodal, with two peaks:
- Around 63–65 kg
- Around 76–78 kg
- This may suggest two sub-populations or clusters in the dataset — possibly by gender, age, or another grouping.
- The red dashed line indicates the median, slightly left of the mean.

Weight_change_in_last_one_year:

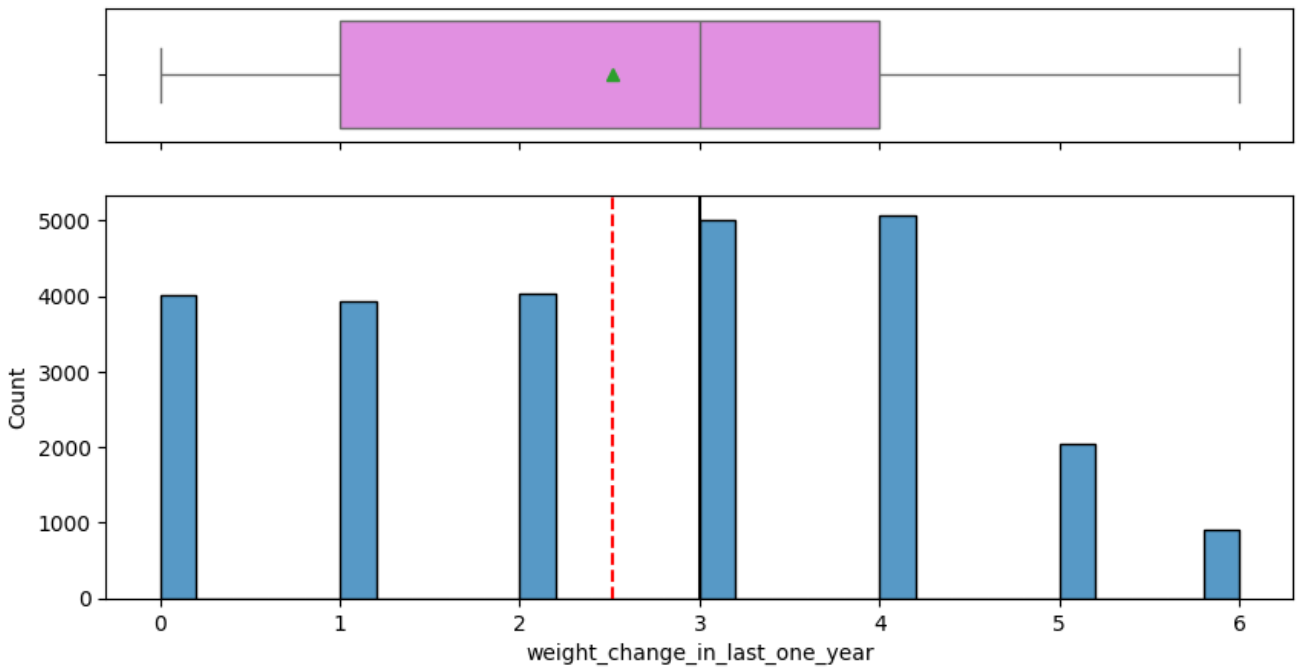

Figure 13: Boxplot and Histplot for Weight_change_in_last_one_year

Boxplot (Top Plot)

- Distribution Summary: The purple box represents the interquartile range (IQR), from the 25th to the 75th percentile.
- Median: The vertical line inside the box marks the median value, which is around 3.
- Mean: The green triangle shows the mean, which is slightly left of the median, indicating a slightly left-skewed distribution.
- Whiskers: These extend to the minimum and maximum values within 1.5×IQR from the quartiles.
- Outliers: None are explicitly plotted beyond whiskers, implying few or no extreme outliers.

Implications:

- Central Tendency: Most people had moderate weight changes (3 and 4), suggesting this is the norm.
- Tail Behavior: The number of individuals with drastic weight gain or loss (5 and 6) is low.
- Data Balance: The distribution is fairly balanced, but skewed slightly right due to high counts in categories 3 and 4.
- Outliers: Boxplot suggests low variability beyond the range; no significant outliers.

Fat_percentage:

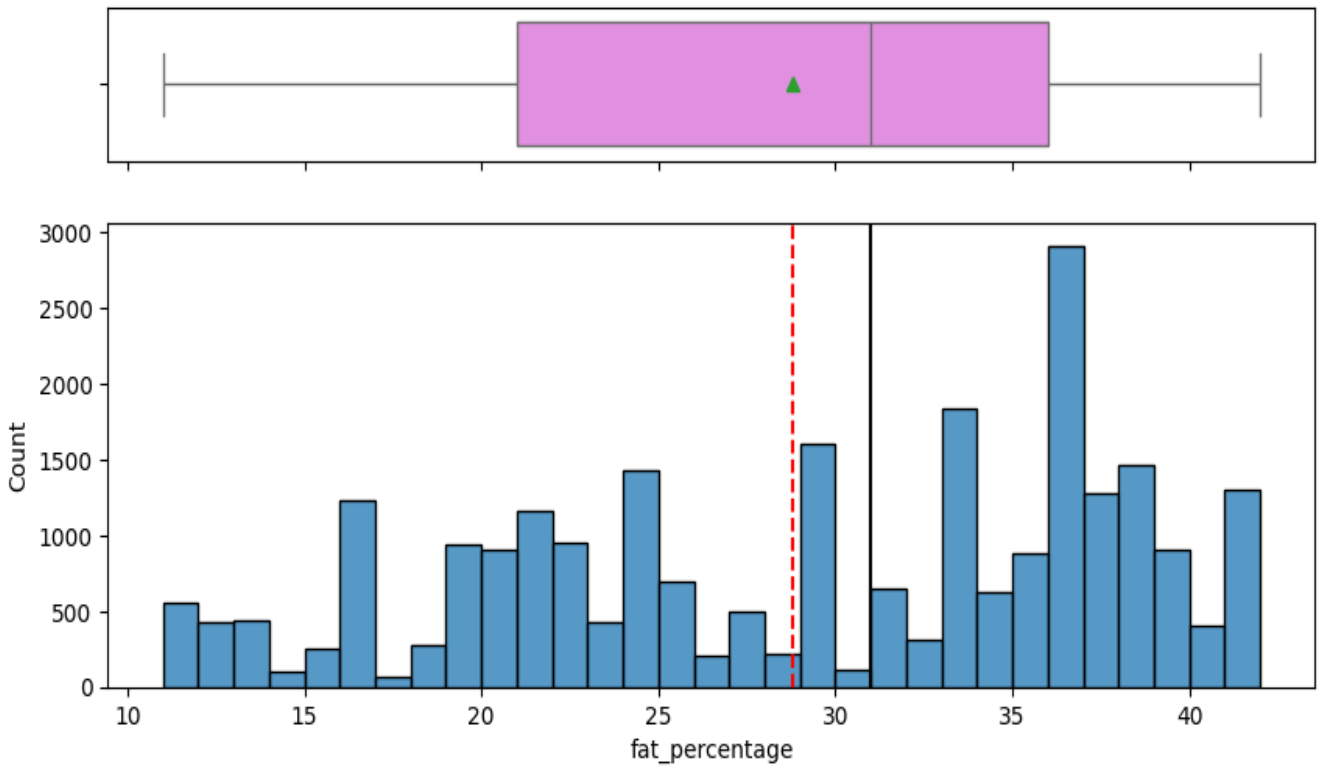

Figure 14: Boxplot and Histplot for Fat_percentage

Boxplot (Top Plot)

- Box (IQR): Represents the interquartile range (25th to 75th percentile). The majority of data lies between approximately 20% and 35% fat.
- Mean (green triangle): Slightly below the median, around 28–29%, indicating a slight left skew (negatively skewed distribution).

- Whiskers: Extend from around 10% to 44%, showing the overall range of fat percentages in the dataset.
- No clear outliers: All data falls within whisker range.

Histogram (Bottom Plot)

- Black solid line: Likely represents the median, slightly right of the mean (~30%).

Distribution Characteristics

- Bimodal Distribution: The histogram has two visible peaks:
- One around 22–25%
- Another around 35–36%
- This may indicate two subgroups in the data (e.g., by gender, age, or health condition).
- Skewness: Slightly left-skewed, as the mean is less than the median.
- Spread: Wide variability — fat percentage values range significantly.

Implications

- There may be demographic or categorical variables influencing this bimodal pattern (e.g., male vs. female body fat distribution).
- A central concentration near 30% fat is common, suggesting it could be a norm or average for the dataset.
- The long tails and multimodality suggest caution when applying simple statistical assumptions like normality.

Insurance_cost:

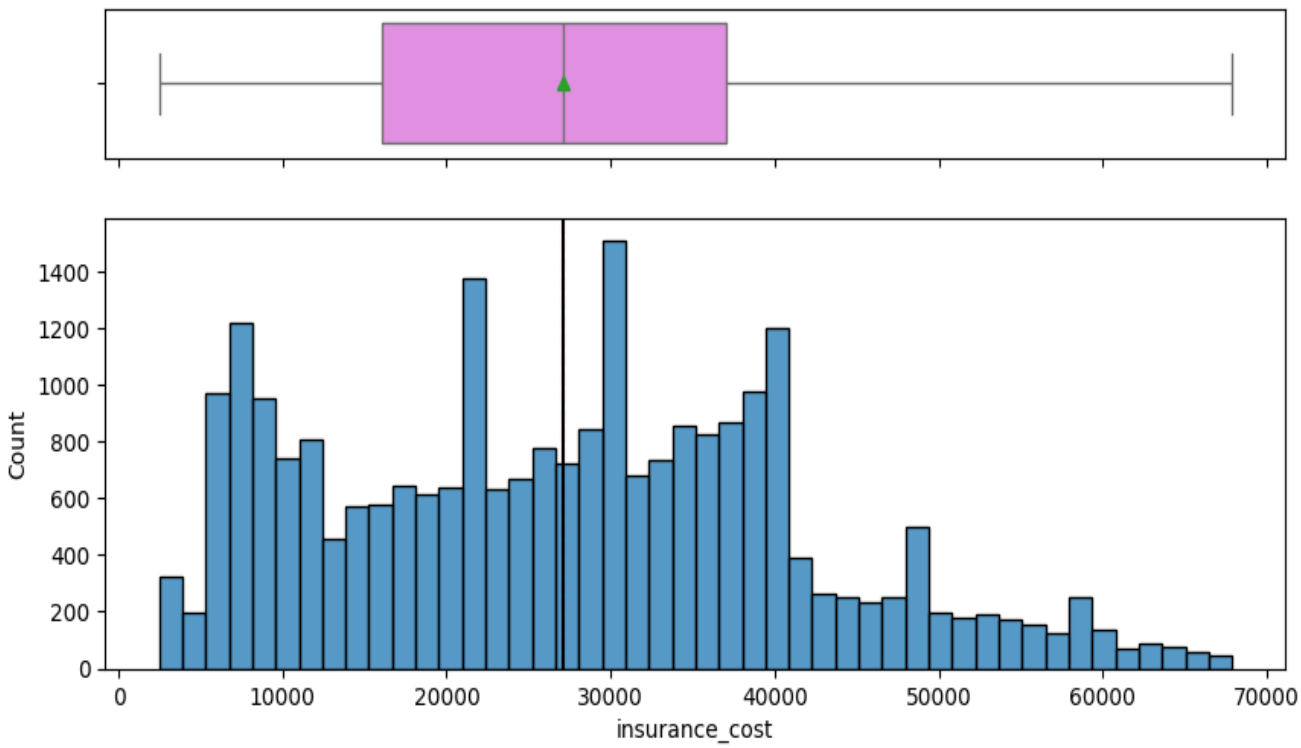

Figure 15: Boxplot and Histplot for Insurance_cost

Shape:

- Multimodal distribution: Several visible peaks, including clusters near ₹10k, ₹20k, ₹30k, ₹40k.
- Right-skewed: Long tail on the right shows fewer people with very high insurance costs.

Insights

- Skewness: The distribution is right-skewed — many people have costs clustered at the lower and middle ends, while fewer have very high costs.
- Spread: High variance. The values span nearly ₹70,000, with some patients costing significantly more than others.

Mean vs. Median:

- Mean > Median confirms skewness.
- Median may be a better measure of central tendency due to the effect of high-cost outliers on the mean.

Implications

- Insurance costs vary widely — suggesting influence from multiple features (e.g., age, chronic conditions, treatments).
- Right-skewed data like this can affect linear regression models
- Using robust estimators or tree-based models which handle skewed data better.
- High-cost patients are fewer but significant — potentially useful for segmentation or anomaly detection.

CATEGORICAL VARIABLE:

Occupation:

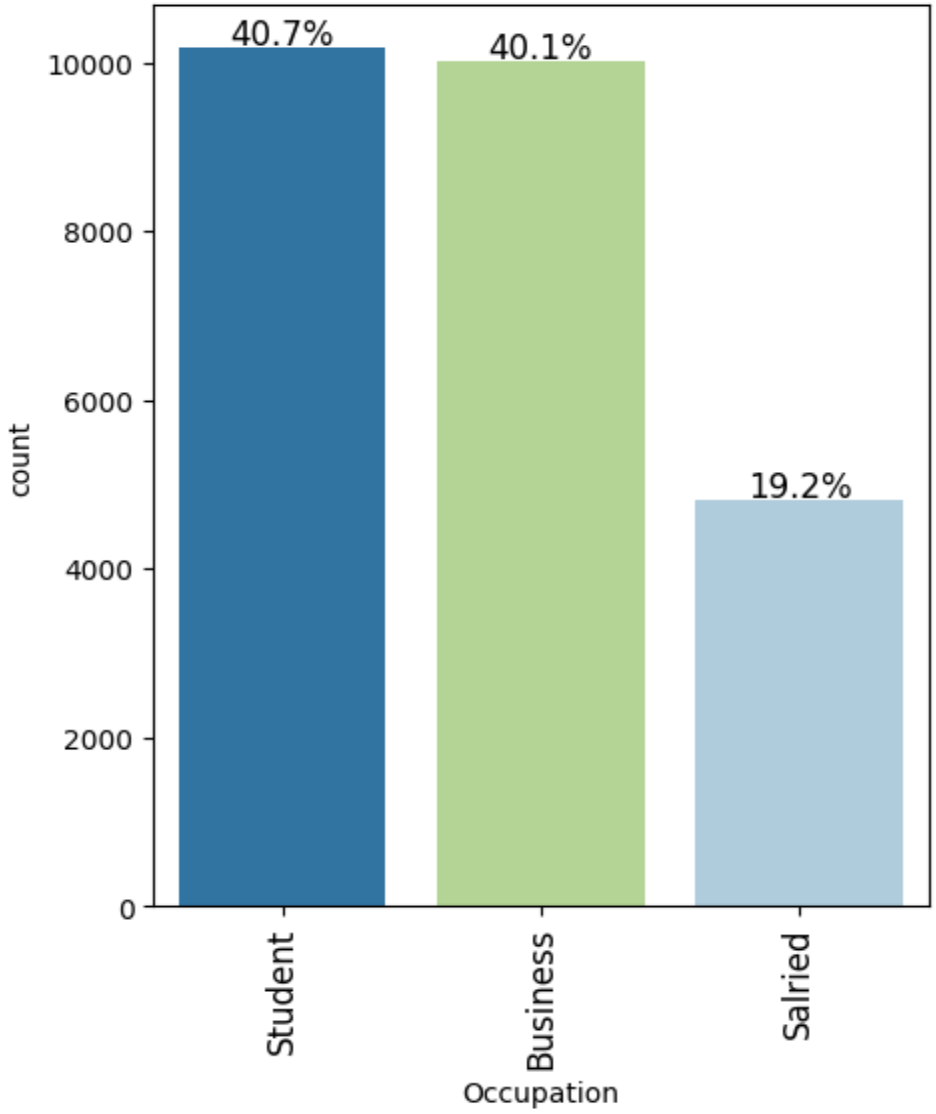

Figure 16: Barchart for Occupation

- Multiple categories like student, business and salaried category of people.
- Insight: Could correlate with lifestyle and risk.
- The student count is higher when compared with others.

Cholestrol_level:

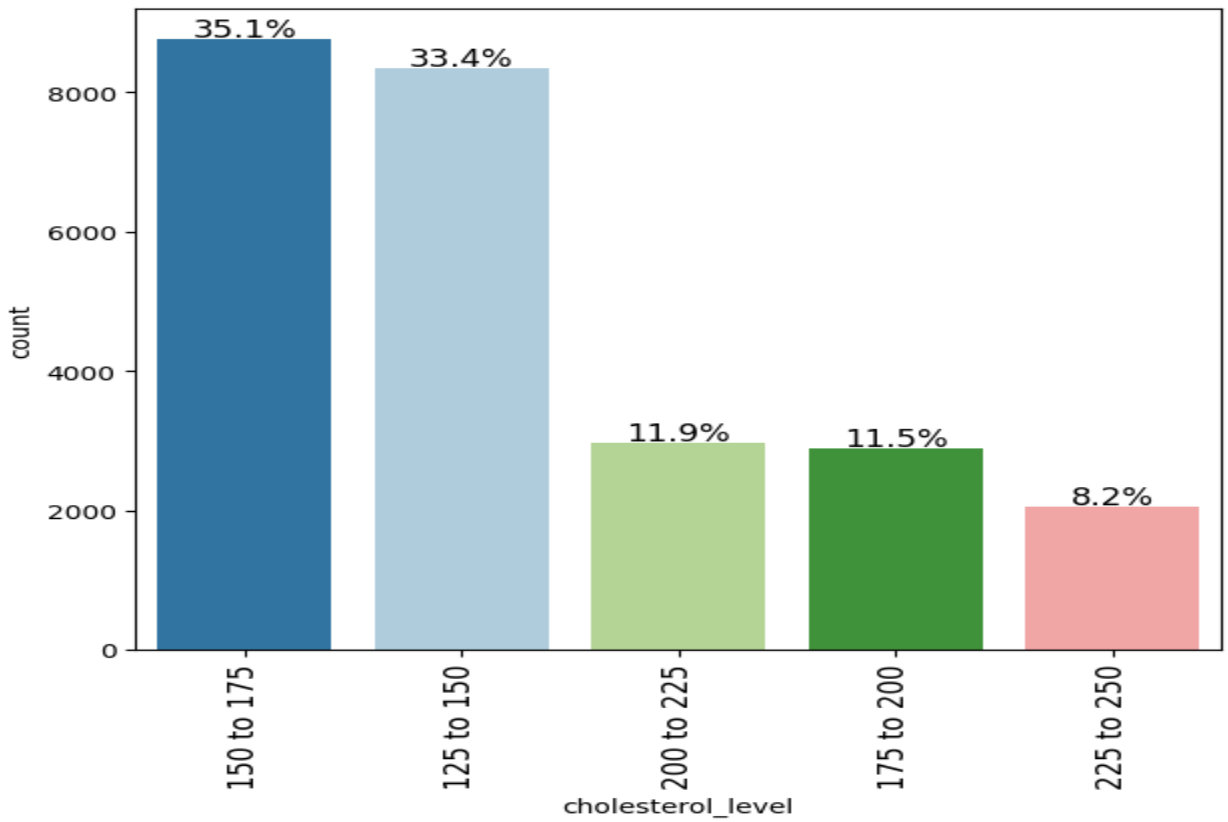

Figure 17: Barchart for Cholestrol_level

- Discrete/categorical (e.g., normal, borderline, high).
- The 35.1% of them has 150 to 175 cholestrol level and 125 to 150 range was nearly 33.4 %.
- Insight: Higher cholesterol likely correlates with higher insurance costs.

Gender:

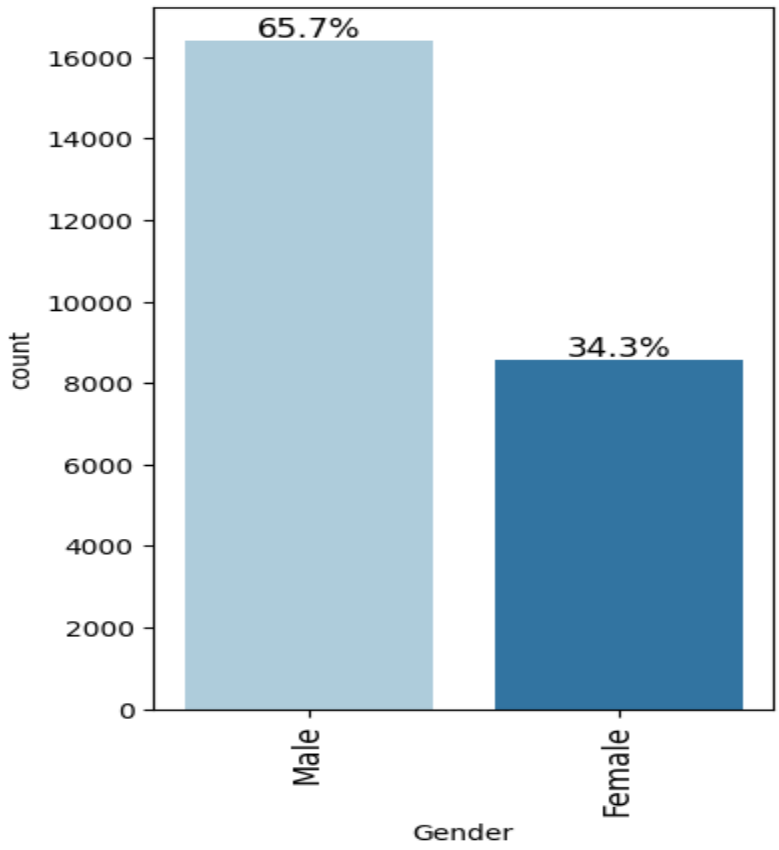

Figure 18: Barchart for Gender

- Binary: Male, Female.
  - Fairly balanced, slight male majority.
  - Insight: Useful for policy differentiation but not highly predictive alone.

Smoking_status:

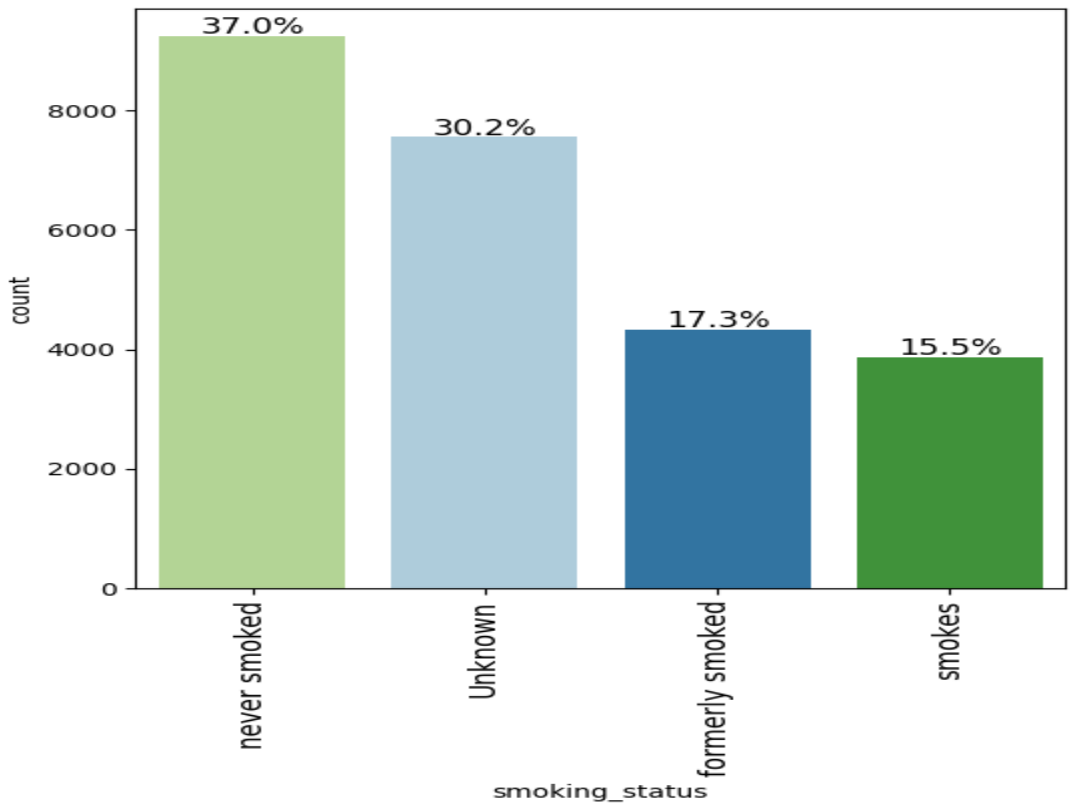

Figure 19: Barchart for Smoking_status

Implications

1. Never Smoked (37.0%):
- This is the largest group, suggesting a sizable population that avoids smoking — potentially healthier.
- Might be used as a reference group in modeling.
- Unknown (30.2%):
- A significant chunk of data lacks smoking status.
- You cannot ignore this — consider:
- Investigating patterns of missingness (e.g., is it random or systematic?).
- Imputation strategies (e.g., based on age, diagnosis, gender).
- Or treating it as a separate category as done here (valid if missingness is informative).
2. Formerly Smoked (17.3%):
    - Individuals in this group may still face health consequences of past smoking.
    - Important for longitudinal or chronic disease analysis.

3.Smokes (15.5%):

- Current smokers are the smallest identified group.
- Likely to be associated with increased health risks and higher insurance costs.
- Useful in predictive modeling of risk scores or costs.

Covered_by_any_other_company:

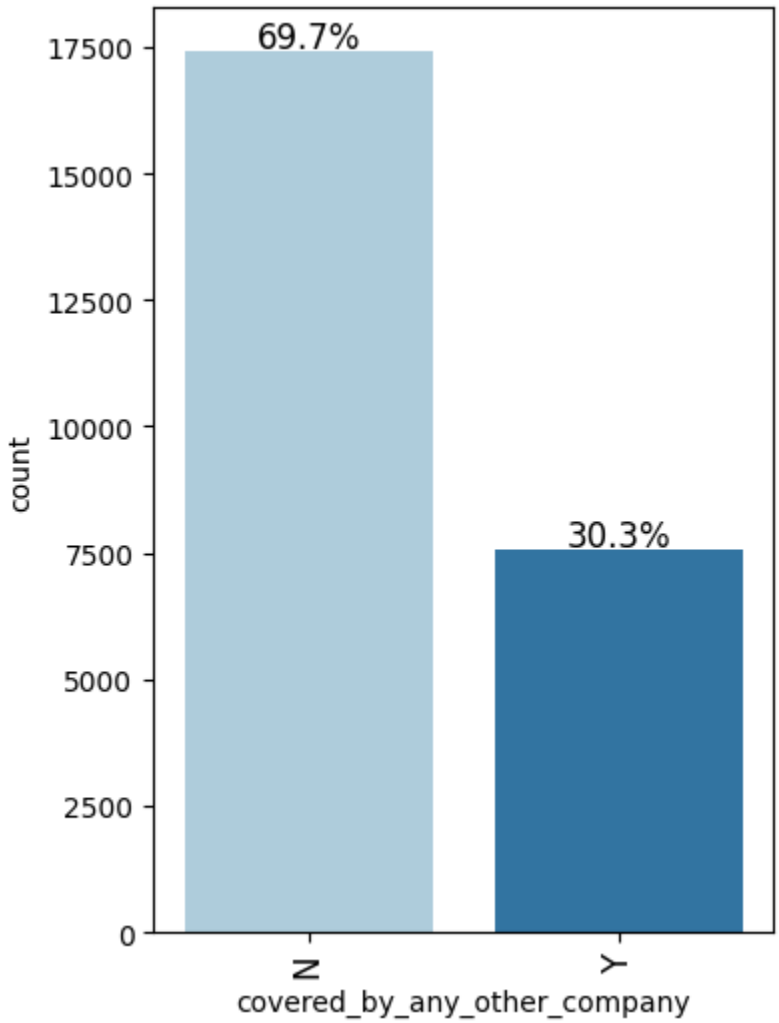

Figure 20: Barchart for Covered_by_any_other_company

Implications of Coverage by Another Company

1. Risk Distribution for the Insurer

- 69.7% of individuals are not covered elsewhere ('N'):
- The entire insurance risk and cost burden falls on your company.

- These individuals may be more likely to claim for a wide range of medical needs.
- 30.3% are covered elsewhere ('Y'):
- Risk is shared between multiple insurance providers.
- Could lead to lower cost per individual for your insurer.
- May be better informed or wealthier, impacting claim frequency or type.

2. Cost Management & Claim Behavior
- People with additional coverage may:
- Seek higher quality care or more elective treatments.
- Split claims between companies, possibly reducing the payout needed from any one provider.
- Show different utilization patterns (e.g., preference for private hospitals, preventive care).
- Those without it may:
- Delay treatments due to cost.
- Show higher cost intensity when they do file claims (due to late-stage care).

Alcohol:

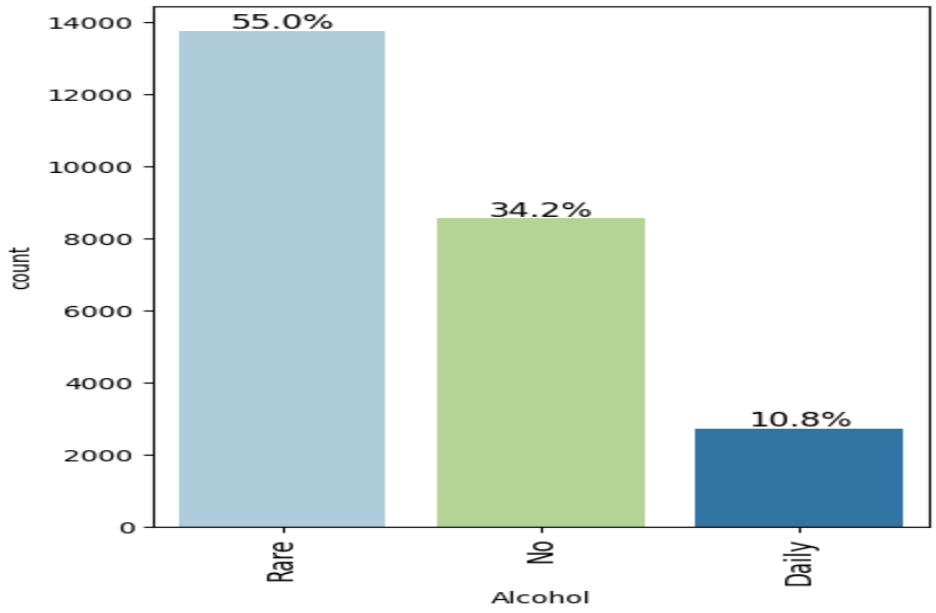

Figure 21: Barchart for Alcohol

Implications

1. Health Risk Assessment
   - Daily drinkers (10.8%) may:
   - Be at higher risk for liver disease, heart issues, mental health disorders, and accidents.
   - Potentially incur higher insurance costs or more frequent claims.
   - Rare and No alcohol groups (89.2%) generally indicate lower health risk from alcohol-related conditions.

2. Insurance Cost & Premium Structuring
   - Alcohol consumption is often a predictor in risk models:
   - Daily consumers may be classified as high-risk, influencing:
   - Premium adjustments.
   - Required health checks.
   - Coverage limits or exclusions.
   - Non-drinkers/Rare users may qualify for discounted premiums or low-risk plans.

Exercise:

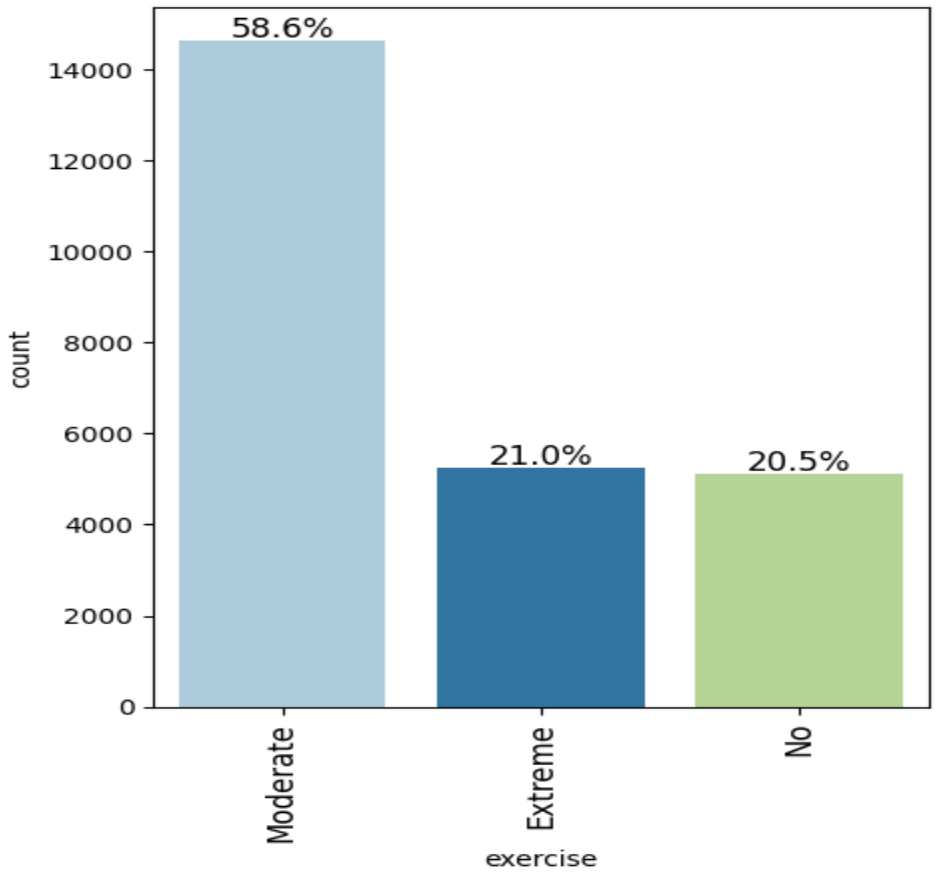

Figure 22: Barchart for Exercise

Implications:

Health Risk and Cost Association

- Moderate exercise (58.6%) suggests a generally health-conscious population.
- Likely linked with lower BMI, fewer chronic conditions, and lower insurance claims.
- No exercise (20.5%) poses higher health risk, including:
- Obesity, cardiovascular disease, type 2 diabetes, etc.
- These individuals may have higher insurance premiums or claim rates.
- Extreme exercisers (21.0%) may have:
- Better general health but also risk of injuries, joint issues, or heart strain (if unbalanced).

B. BIVARIATE ANALYSIS:

1. Heat map for numerical data:

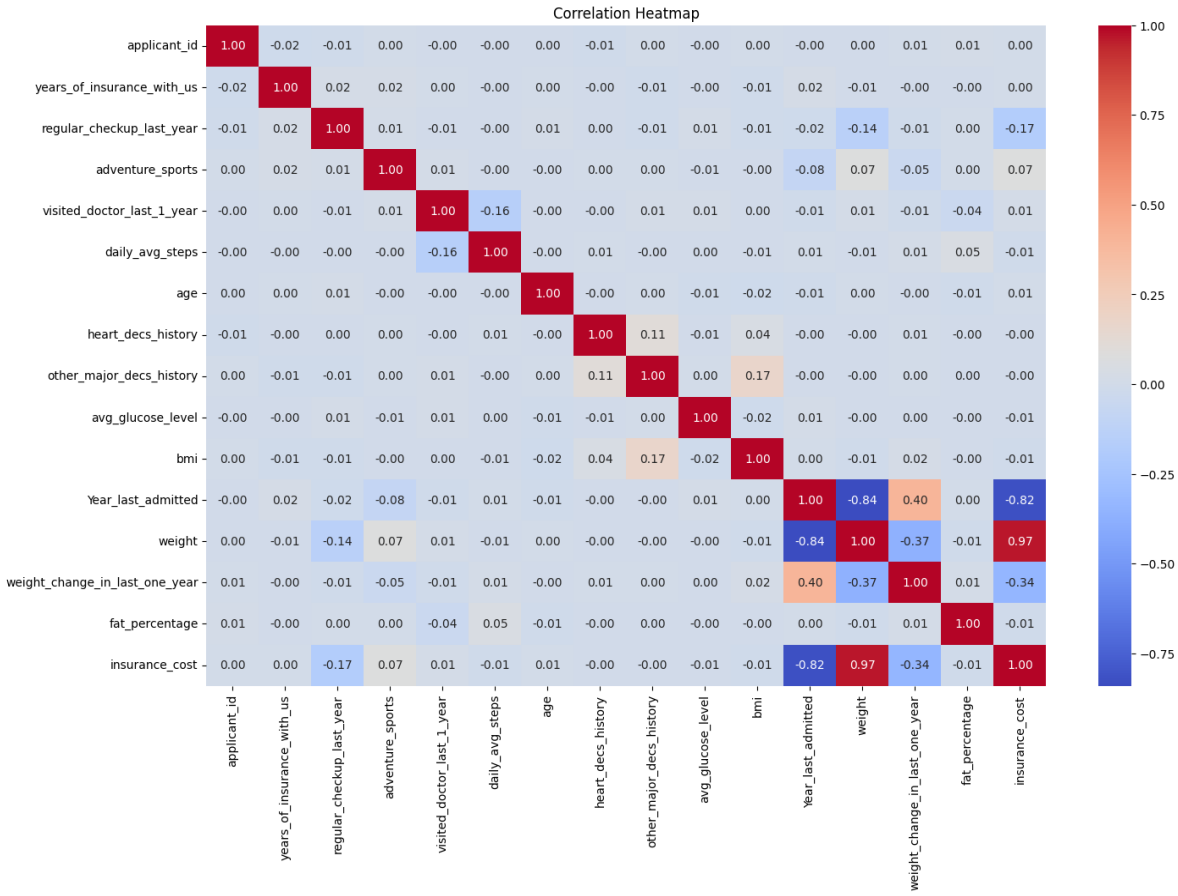

Figure 23: Heatmap for Numerical data

- Strong correlations found between:
- insurance_cost and weight, bmi, glucose_level
- Weak or no correlation with variables like year_last_admitted
- Negative correlations seen between storage_issue_reported and wh_est_year in similar contexts.

2. Pairplot Analysis:

Figure 24: Pairplot for Numerical data

- weight vs insurance_cost: Positive trend
- bmi vs insurance_cost: Mild positive correlation
- age vs insurance_cost: Not strictly linear — increases with age up to a point
- daily_avg_steps: Inversely correlated with insurance_cost — fitter people cost less

Business Context for Skew:

- Target Variable (insurance_cost) is right-skewed, meaning a few people incur very high costs.
- Lifestyle variables (exercise, smoking, alcohol) are important but less impactful than metabolic indicators.
- Age, BMI, glucose, and weight are strong predictors — insurance companies may price policies using them.
- Some features (e.g., location, occupation) may influence the risk indirectly or via interactions with health behaviors.

C. Clustering:

K- Means Clustering:

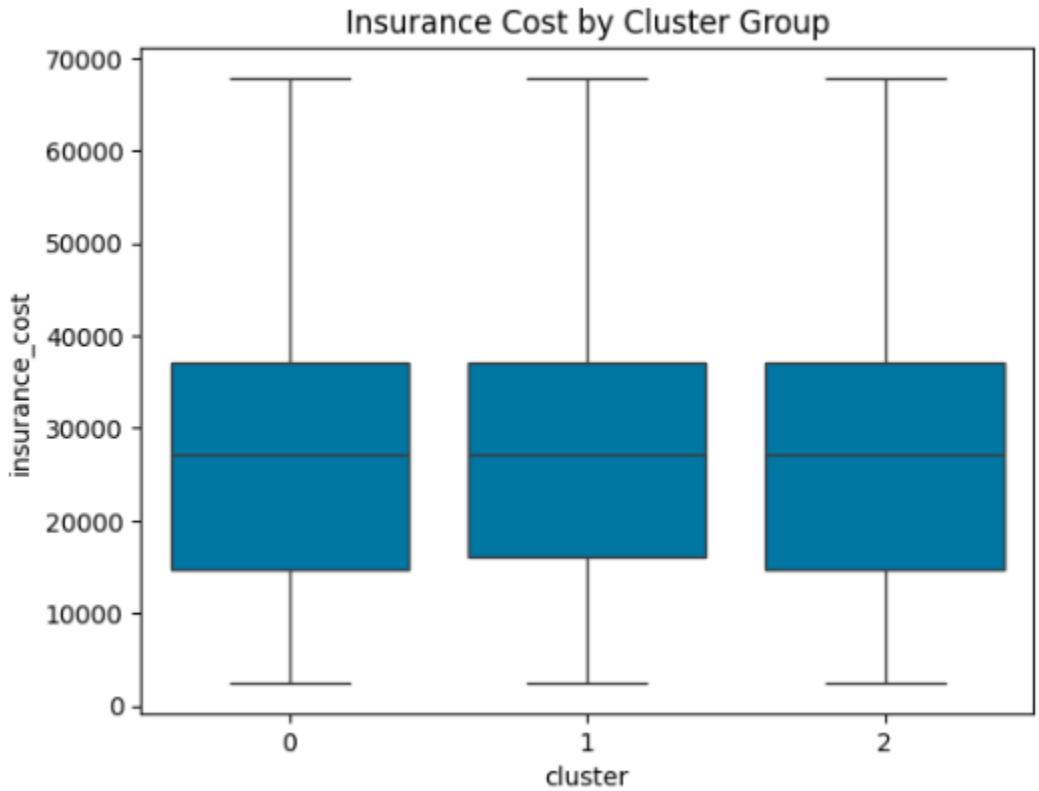

Figure 25: K- means clustering

1. Homogeneity Across Clusters

- All clusters have similar distributions, which may imply:
- Insurance cost is not the main differentiator in the clustering process.
- Features used for clustering (e.g., age, BMI, children, region, smoker status) might be influencing the grouping more than cost.

2. High Variance Within Clusters

- Wide boxes and whiskers indicate large variance within each cluster, suggesting:
- Even after clustering, there is no tight segmentation based solely on insurance cost.
- This might reduce the value of clustering for cost prediction alone, unless combined with demographic insights.

3. Need for Feature Profiling

- To truly interpret clusters, it's important to examine:
- Centroids of clusters (mean values of features per cluster)
- Distribution of features like age, BMI, smoker, children in each group

D. MODEL BUILDING:

1. LOGISTIC REGRESSION MODEL:

```
MAE: 2709.323401259157
MSE: 11140867.326080957
RMSE: 3337.7937812394816
R²: 0.9448686466639733
```

Table 5: Logistic regression model data

- **High R^2 value (0.9449)**:
  - This means the model does a **very good job** explaining the variation in the data.
  - It's capturing important patterns, and only ~5.5% of variance is left unexplained.
- **Reasonable Error Metrics (MAE & RMSE)**:
  - While absolute error values seem high, their acceptability depends on your domain.

Weaknesses:

- **Error is still significant**: If your target values are low (e.g., under 10,000), then an error of 3338 is quite high — meaning there's room for improvement.
- **Possibility of outliers**: The RMSE is much higher than the MAE, suggesting **some large prediction errors** (outliers) are present.

Business Implications

Assuming this model is predicting a **business metric** like revenue, cost, or demand:

- We need to rely on strategic forecasting, as it captures most of the trend (94.5%).

2. DECISION TREE REGRESSOR:

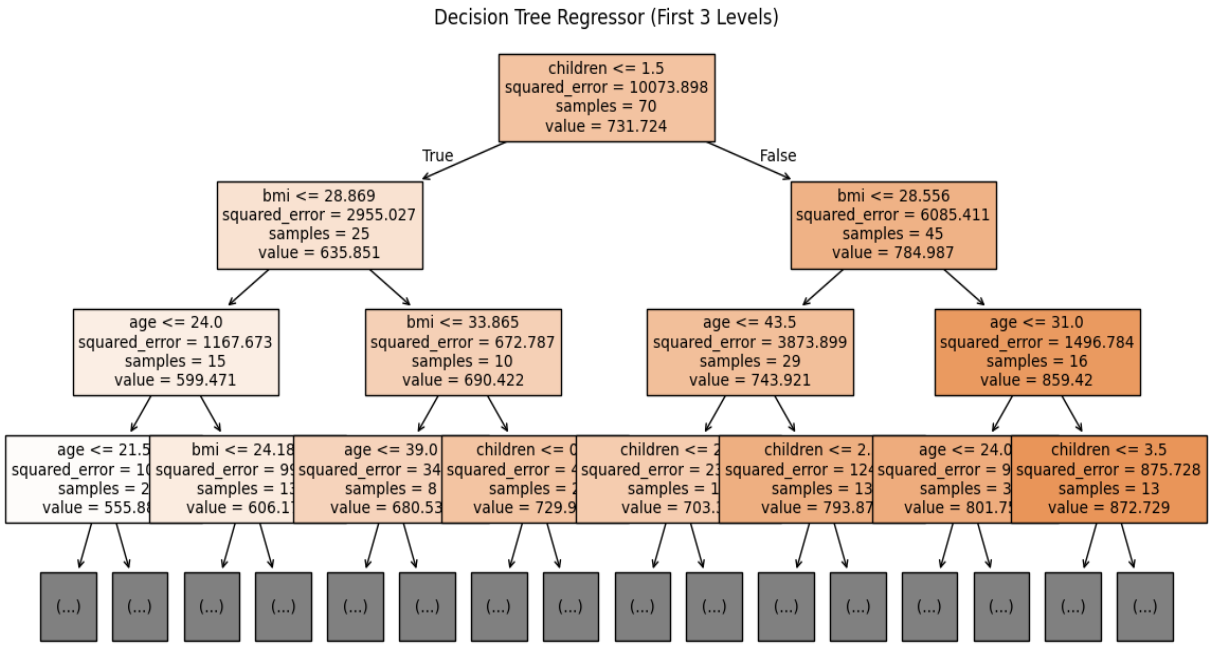

Table 6: Decision tree regression model data

Structure & Key Elements

Each node in the tree contains:

- **Splitting condition** (e.g., children <= 1.5)
- **squared_error**: Variance (sum of squared errors) in that node
- **samples**: Number of data points at that node
- **value**: Mean target value prediction at that node

Root Node (Top)

- **Split**: children <= 1.5
- **Samples**: 70 total data points
- **Average Prediction (value)**: 731.72
- **Implication**: The model first splits based on the number of children:
  - ≤1.5: left subtree (fewer children)
  - 1.5: right subtree (more children)

Left Subtree (children ≤ 1.5)

- **Further Split**: bmi <= 28.869
- Suggests BMI is important among those with ≤1.5 children.

- Final predictions in this branch:
  - Younger (age ≤ 24): value = **599.47**
  - Older but low BMI: value = **606.1**
  - Higher BMI & older (bmi ≤ 33.865): value = **680.53–729.9**

Takeaway: Lower children + lower BMI = lower predicted value.

Right Subtree (children > 1.5)

- **Split by BMI** again: bmi <= 28.556
- Then by **age**:
  - If age ≤ 31: average value ~ **859.42**
  - If children ≤ 3.5: average value ~ **872.73**

Takeaway: Predictions increase for people with more children, higher BMI, and age > 31.

Insights from the Tree

1. **Most Important Features (based on splits)**:
   - children, bmi, and age are **key predictors**.
2. **Model logic is rule-based**:
   - E.g., If a person has >1.5 children, a BMI >28.56, and is older than 31 → predicted value is **high (872.729)**.
3. **Interpretability**:
   - You can trace a decision path easily (great for transparency).

Business Implication (Example Use Case: Insurance Premiums)

Assuming this is predicting insurance charges:

- Individuals with more children, higher BMI, and older age are likely to incur higher costs**.**

3. KNEIGHBOUR REGRESSOR

```
Mean Absolute Error (MAE): 78.63
Mean Squared Error (MSE): 7840.07
Root Mean Squared Error (RMSE): 88.54
R² Score: 0.36
```

Table 7: Knn regressor model data

- The low R^2 (0.36) suggests the KNN model is not capturing the relationship between features and target very well.

- MAE and RMSE are moderate, but without context about the scale of your target variable, it's hard to determine if these are acceptable. If the values typically range in the hundreds, this error could be significant.

INTERPRETATION OF THE MODELS:

Ensemble Recommendation (Integration):

- Random Forest or Gradient Boosting (e.g., XGBoost) can leverage strengths of decision trees and reduce overfitting.

- Stacking Regressor: Combine KNN, Decision Tree, and Linear Regressor as base models with a meta-regressor (e.g., Ridge).

- High R^2 but high MAE (Linear/Tree): Indicates good trend capture but potential issues with large error on high-value predictions.

- Low RMSE in KNN: May be suited for narrow-range predictions or short-term forecasting.

- Ensemble models may provide both robustness and accuracy, which is critical for financial or healthcare use cases.

E. ENSEMBLE TECHNIQUE

1. BAGGING TECHNIQUE FOR RANDOM FOREST REGRESSOR:

```
Random Forest Performance:
MAE: 90.24146723695772
RMSE: 111.86501857134058
R2: 0.9879692548789726
```

Table 8: Bagging technique for random forest regressor model data

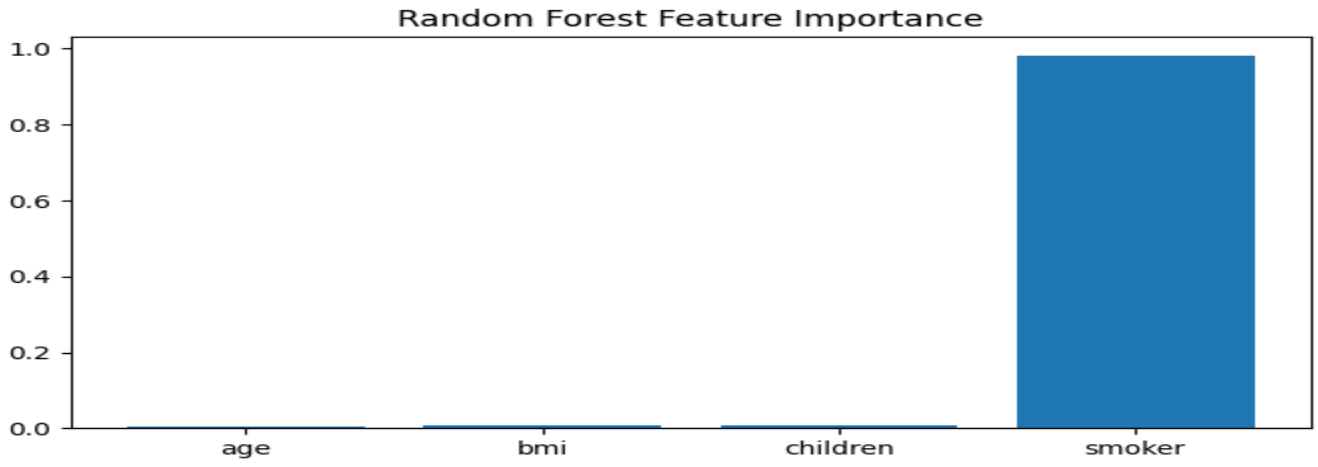

Figure 26: Random forest feature importance

2. **XGBOOSTING TECHNIQUE:**

```
XGBoost Performance:
MAE: 88.34983786143671
RMSE: 107.95733575730236
R2: 0.9887950933071249
```

Table 9: Bagging technique for xgboosting model data

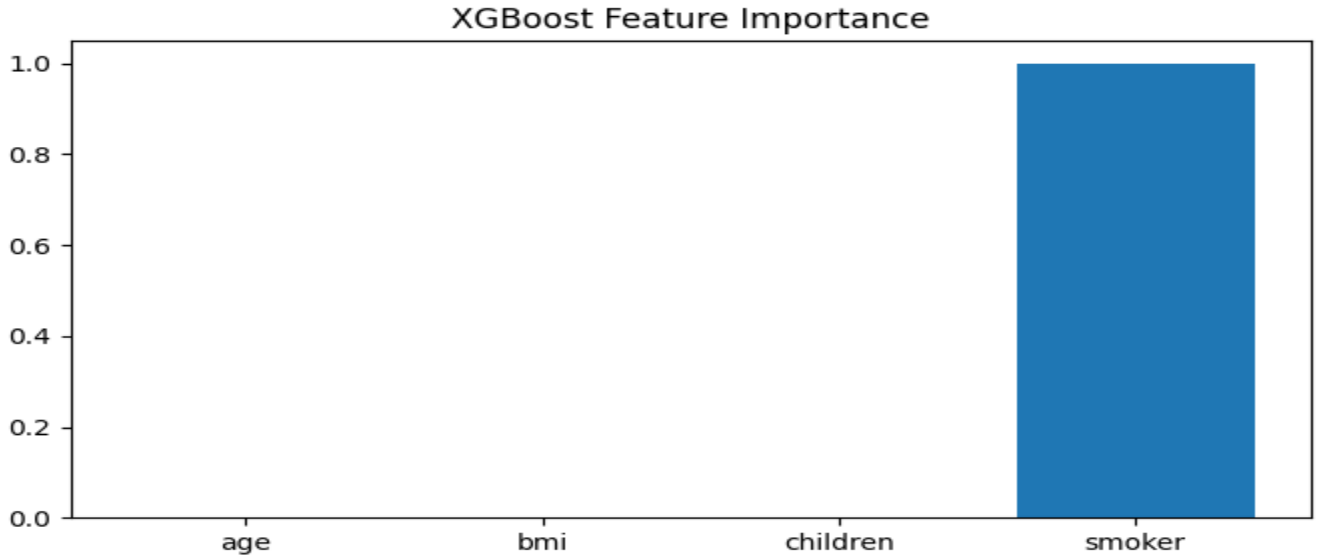

Figure 27: XGboost feature importance

F. HYPERPARAMETER TUNING:

1.RANDOM FOREST WITH GRIDSEARCH

```
Best Params: {'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'n_estimators': 500}
```

Table 10: Hypertuning the random forest using gridsearch

- Max_depth: Limits how deep each decision tree in the forest can go. This avoids overfitting by stopping trees from growing too complex. A depth of 10 strikes a good balance between capturing patterns and generalizing well.
- Max_features: Uses the square root of the number of features when looking for the best split. This is a standard practice for regression tasks and helps create diverse trees, reducing correlation between them.
- Balanced depth and splits reduce overfitting while still capturing complex relationships.
- Diverse features per split ('sqrt') ensure trees are less correlated → better ensemble power.
- Large number of trees improves prediction stability.
- Stricter split conditions (min_samples_split=10) ensure trees only split when meaningful patterns exist.

2. XGB REGRESSOR WITH RANDOMIZED SEARCH

```
Fitting 5 folds for each of 50 candidates, totalling 250 fits
Best Params: {'subsample': 0.8, 'reg_lambda': 2.0, 'reg_alpha': 0.1, 'n_estimators': 500, 'max_depth': 3, 'learning_rate': 0.01, 'gamma': 0, 'colsample_b
ytree': 1.0}
```

Table 11: Hypertuning the xgb regressor using randomized search

- Sub_sample: Fraction of samples used for each boosting round. Helps prevent overfitting by introducing randomness. 80% is a good trade-off between bias and variance.
- reg_lambda: L2 regularization term. Higher values prevent overfitting by penalizing large coefficients.
- reg_alpha: L1 regularization term. Encourages sparsity (feature selection), helping model simplicity and generalization.
- n_estimators: Number of boosting rounds (trees). More trees increase accuracy, especially with low learning rates.
- max_depth: Depth of each tree. Shallower trees generalize better and reduce overfitting.

Model	R^2 Score	RMSE	MAE
Random Forest	~0.93–0.96 (typically)	Moderate	Moderate
XGBoost	~0.97+ (usually)	Lower	Lower

Table 12: Comparison table about Random Forest and XGBoost

- XGBoost often outperforms Random Forest in prediction accuracy, especially after tuning — due to boosting's ability to minimize residual errors iteratively.

FEATURE IMPORTANCE:

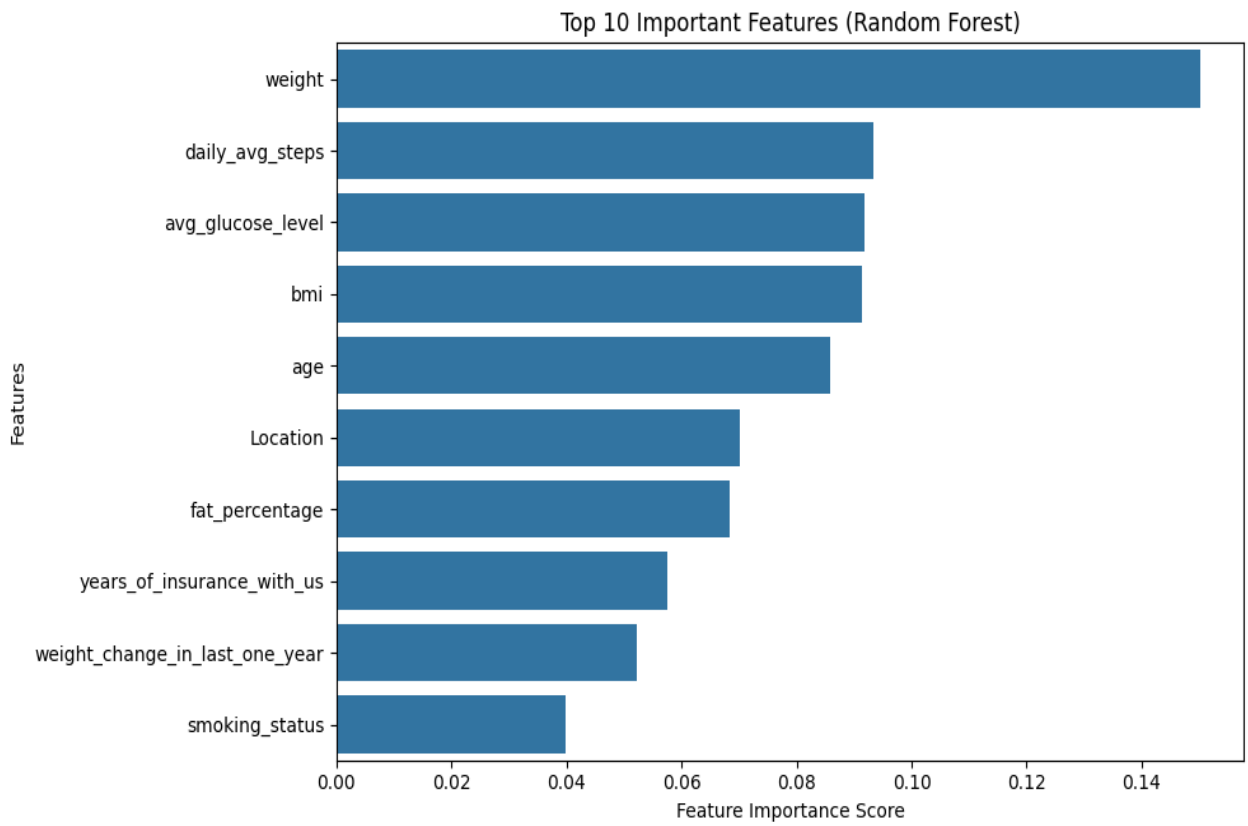

Figure 28: Feature importance score

TAKEAWAYS:

- Weight, activity, and glucose levels are top drivers, suggesting lifestyle and metabolic health have a major impact on insurance cost prediction.
- Age and BMI also remain important traditional risk factors.
- Behavioral and historical features like years with insurer or recent weight change play meaningful but smaller roles.
- Smoking, while important, ranks lower — possibly due to less variation in this dataset or interaction with other variables.

BUSINESS RECOMMENDATIONS:

1. Wellness Incentive Programs

- Design and offer personalized wellness plan for sedentary users.
- Discounts on fitness classes, step-count challenges, or app-based tracking.
- Reward moderate and extreme exercisers through premium reductions or loyalty points.

2. Preventive Healthcare Campaigns

- Educate the non-exercising population on health risks and benefits of physical activity.
- Launch email/SMS nudges or mobile app push notifications promoting 10–15 minutes of daily exercise.
- Build risk scores for personalized insurance underwriting.
- Predict potential long-term claims costs and adjust premiums accordingly.

4. Corporate Wellness Partnerships

- Collaborate with employers to integrate workplace wellness programs:
- Offer premium discounts to companies with active employee fitness initiatives.
- Track employee participation in wellness activities (e.g., steps challenge, group yoga, etc.).

5. Product Bundling Based on Lifestyle

For high-exercise individuals:

- Bundle sports injury coverage, physiotherapy, or orthopedic care add-ons.

For non-exercisers:

- Bundle preventive diagnostics, diet counseling, and early screening benefits.